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Health Insurance Cost & Claims Analytics (Project HIC-2026-001)
Stage 2 of 3: Exploratory Data Analysis (EDA)

Extracted directly from the project's analysis notebook (analysis.html).
Code is reproduced exactly as authored -- nothing has been changed,
only separated into its own file. Cell boundaries from the original
notebook are kept as "# %%" markers, which VS Code and Jupyter both
recognize as runnable cells.

Run this after 01_data_cleaning.py. It assumes the cleaned `df`
(and the standard imports/settings) from that stage are already
available in the session/kernel.
"""



# EDA — distributions


numeric_cols = ["age", "bmi", "bloodpressure", "children", "claim"]
df[numeric_cols].hist(bins=20, figsize=(12, 8), color="skyblue", edgecolor="blue")
plt.suptitle("Distribution of Numerical Features", fontsize=16)
plt.tight_layout()
plt.show()

cat_cols = ["gender", "diabetic", "smoker", "region"]
plt.figure(figsize=(12, 8))
for i, col in enumerate(cat_cols, 1):
    plt.subplot(2, 2, i)
    sns.countplot(data=df, x=col)
    plt.title(f"Distribution of {col}")
plt.tight_layout()
plt.show()


# Calculates the average insurance claim amount grouped by gender and smoker status, rounded to 2
decimal places.


df.groupby(["gender", "smoker"])["claim"].mean().round(2)


plt.figure(figsize=(8,5))
sns.barplot(data=df, x="gender", y="claim", hue="smoker", estimator="mean", errorbar="sd")
plt.title("Average Insurance Claim by Gender and Smoking Status")
plt.show()

# %%
# EDA — claim by smoker / diabetic / region

# %%
print(df.groupby(["gender", "smoker"])["claim"].mean().round(2))

pivot_region_diabetic = df.groupby(["region", "diabetic"])["claim"].mean().unstack()
print(pivot_region_diabetic)

pivot_region_smoker = pd.pivot_table(df, values="claim", index="region",
                                      columns="smoker", aggfunc="mean")
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print(pivot_region_smoker)

# %%
# Visualizes the pivot table as a grouped bar chart, comparing average claim amounts across
regions, split by diabetic status.


pivot_region_diabetic.plot(kind="bar", figsize=(8,5))
plt.title("Average Claim by Region & Diabetic Status")
plt.ylabel("Mean Claim")
plt.show()


# Interaction check called out in the report: diabetic status alone is
# a weak signal — look at it combined with smoker and BMI category


df["bmi_category_tmp"] = pd.cut(
    df["bmi"], bins=[0, 18.5, 24.9, 29.9, 100],
    labels=["underweight", "normal", "overweight", "obese"]
)
print(df.groupby(["diabetic", "bmi_category_tmp"])["claim"].mean().round(2))
print(df.groupby(["diabetic", "smoker"])["claim"].mean().round(2))
df = df.drop(columns=["bmi_category_tmp"])


# Correlation heatmap


plt.figure(figsize=(8, 6))
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()


# Visualizes the relationship between age and claim amount as a scatter plot, with color showing
smoker status and marker shape showing gender.


sns.scatterplot(data=df, x="age", y="claim", hue="smoker", style="gender")
plt.title("Claim vs Age by Smoker & Gender")
plt.show()


# bmi and claim


sns.regplot(data=df, x="bmi", y="claim", scatter_kws={"alpha": 0.6})
plt.title("Relationship between BMI & Claim Amount")
plt.show()


# Visualizes the spread and distribution of claim amounts for each number of children (using a
boxplot to show median, quartiles, and outliers).

# %%
sns.boxplot(data =df, x="children", y = "claim")
plt.title("claim distribution by children")
plt.show()

# %%
# Age group / BMI category for reporting
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# %%
df["age_group"] = pd.cut(df["age"], bins=[0, 18, 30, 45, 60, 110],
                          labels=["<18", "18-30", "31-45", "46-60", "60+"])
print(df["age_group"].value_counts())

df["bmi_category"] = pd.cut(df["bmi"], bins=[0, 18.5, 24.9, 29.9, 100],
                             labels=["underweight", "normal", "overweight", "obese"])
print(df["bmi_category"].value_counts())


sns.boxplot(data=df, x="bmi_category", y="claim")
plt.title("Claim Distribution by BMI Category")
plt.show()


sns.barplot(data = df, x="age_group",y ="claim",estimator ="mean",errorbar ="sd")
plt.title("average claim by age group")
plt.show()





# Volume-vs-severity split flagged in the report: is the 31-45 / Southeast
# concentration a volume effect or a severity effect?


print(df.groupby("age_group")["claim"].agg(["count", "mean", "sum"]))
print(df.groupby("region")["claim"].agg(["count", "mean", "sum"]))


# Regional Summary: Smoker Rate & Average Claim
# Groups the data by region to calculate two key metrics: the percentage of smokers in each
region, and the average claim amount per region.


region_stats = df.groupby("region").agg(
    smoker_rate=("smoker", lambda x: (x == "yes").mean() * 100),
    mean_claim=("claim", "mean")
).reset_index()


region_stats


fig, ax1 = plt.subplots(figsize=(8,5))

sns.barplot(data=region_stats, x="region", y="smoker_rate", ax=ax1, alpha=0.6)

ax2 = ax1.twinx()
sns.lineplot(data=region_stats, x="region", y="mean_claim", ax=ax2, color="red", marker="o")

ax1.set_ylabel("Smoker Rate (%)")
ax2.set_ylabel("Average Claim ($)")
plt.title("Smoker Rate and Average Claim by Region")
plt.show()
```